TEACHING STATEMENT

LUCAS R. KERBS

I think that my job as an educator and as a researcher is to make math available to those around me. When education is done, well it is accessible, engaging, and inspiring. When education is done poorly, it is opaque, boring, and disinterested. Most students leave their university with fond memories of one or two classes that inspired them; classes where they connected with the ideas presented to them and explored them on their own. Unfortunately, a math class is rarely on that list.

Ultimately, education is about asking and answering questions. Knowledge may be handed down from on high during lectures, but *learning* requires students to interface with those ideas on their own, to get their hands dirty with the material and truly engage with it. As process of mathematics is one of building knowledge through asking and answering question, a math class is a prime opportunity for students to engage with both the *process* of education and with the material itself. My time as an instructor during graduate school has been a wonderful testing grounds for my pedagogical philosophy.

During my masters, I was the instructor of record for seven courses—a mixture of precalculus and business calculus. As there was no "master instructor," I was in charge of the entire course, from writing the syllabus and course planning to writing exams and assigning final grades. Through teaching precalculus, I learned that there is a large gap between explaining a concept to students and *teaching* it to them. Definitions are not meant to be introduced as a black box. Instead, definitions are something we choose and we chose those definitions for a specific reason; when students understand this they are able to more easily move from rote memorization to an intuitive understanding.

PhD students at Washington University often have limited or no opportunity to teach a full course on their own—our normal teaching duties are grading and leading recitation. As I believe that instruction is a core part of being a mathematician, I sought out the opportunity to further to refine my pedagogical skill—I spent my first two summers teaching Quantitative Reasoning. As QR was not a prerequisite for any other course and there was only one section every summer, I had a significant amount of freedom with what I chose to cover. These classes both less than ten students with a wide range of mathematical ability—some students has already completed the calculus sequence while others were slated to take college algebra in the fall. As such, I focused the course on building an intuition for the kinds of numbers they would meet in their daily lives. Prior to the course beginning, I intentionally left the last two weeks of class without a plan for content. Instead, I offered students several options and I let their curiosity throughout the semester guide the directions of the lectures. As they had a hand in the course direction, they were bought-in and the lectures centered on collaboratively exploring ideas instead of just explaining the concepts.

In the recitation sections that I led at WashU, I gained a new appreciation for collaborative learning. While the bulk of this experience was helping students with weekly worksheets, I

made a point to make sure that students were always working in groups. Many students are hesitant to ask questions, but giving them a handful of other students to work alongside made those students much more confident asking questions of their peers. Students at different levels of understanding working together brought the idea of education-as-a-process to the surface of the classroom. As a result, when I would visit each group I was able field questions from all the students and I could take the whole group through an idea at the same time. Creating the atmosphere of collaboration shifted the focus of the class; instead of each student facing a worksheet to complete on their own, recitation turned into to a space where all the students worked together to build a collective understanding.

I believe that, in its most pure form, education necessarily involves both learning and teaching—where the two inform one another and make the other better as a result. The process of learning is a two-way street. I expect my students to work hard and lift mathematical weights they may not feel ready for, but in exchange my students can expect me to arrive well prepared to assist them; to answer their questions, ask questions back, and guide them to true understanding.

Department of Mathematics, Washington University in St. Louis, St. Louis, MO

Email address: kerbs@wustl.edu